Hamza Yousaf

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Education

Toronto Metropolitan University — Toronto, ON
Bachelor of Science (Hons) in Computer Science, Co-op — *Expected Graduation: April 2028*

Experience

Software Engineering Intern — July 2026 – September 2026
Expense Trend — *Remote*

- Developed an analytics dashboard for an iOS onboarding A/B testing platform using Next.js and TypeScript.
- Consumed REST APIs to fetch and visualize performance metrics through interactive charts and tables.
- Queried conversion data from a Postgres database using SQL and transformed it into inputs for Google Gemini.
- Engineered an AI service using Node.js and Express to optimize paywall screens based on historical performance.
- Wrote Jest test files for automated backend testing, covering end-to-end integration, edge cases, and rate limiting.

Coding Instructor Volunteer — July 2023 – July 2023
Code Ninjas — *Brampton, ON*

- Delivered lessons in Python, algorithms, problem solving and computational thinking to 30+ students.
- Guided hands on labs with Arduino micro-controllers, schematic design and embedded programming.
- Mentored robotics workshops, improving students critical thinking, problem solving and debugging abilities.

Projects

FarmLite | *JavaScript, React.js, Next.js, Node.js, Express.js, Tailwind, GPT-4 Mini, REST API* | GitHub

- Built a full stack web application which evaluates vegetation fertility using NDVI soil data and LLM analysis.
- Ensured data reliability by cleaning unprocessed coordinate inputs before API requests, reducing failure to <5%.
- Reduced API response time by 60% through data pipelines and payload minimization, measured via Postman.

StudySprout | *JavaScript, MongoDB, React, Express, Node.js, Tailwind CSS, Zustand, Chakra UI* | GitHub

- Built a study tracking platform enabling users to create study goals, track learning hours, and monitor progress.
- Increased backend performance by 35% by refactoring Express middleware and optimizing schema design.
- Improved security by implementing JWT authentication and securing HTTP routes against unauthorized access.

Garbage Classification CNN | *Tensorflow, Matplotlib, NumPy, Python, openCV, JupyterLab* | GitHub

- Improved image classification accuracy by 28% with a convolutional neural network sorting waste into 6 classes.
- Architected a machine learning pipeline including data preprocessing, feature extraction, ReLU and validation.
- Evaluated model generalization by testing predictions on live webcam images, identifying performance limitations.

Planetary Gravity Simulation | *Java, Java Swing* | GitHub

- Architected a planetary gravitational simulation modeling multi-body celestial interactions via Newtonian physics.
- Reduced codebase complexity by restructuring files into modular object-oriented classes for scalability.
- Designed a Java Swing interface allowing users to add, remove, and manipulate celestial bodies in real time.

Extracurriculars

Jean Augustine Hackathon | *C++*

- Developed an automated booking system to manage 50+ customer records, reducing manual entry time by 83%.
- Provided a system to increase customer retention through the implementation of a calculated callback system.
- Generated 120+ service report files for admins and customers to improve record keeping and receipts.

TerraHacks Hackathon | *Django, HTML, CSS, Python*

- Led a team of 3 to deliver survey based eco recovery features with HTML/CSS to ship a functional web platform.
- Refactored backend Django logic to reduce unnecessary database queries and improve API efficiency.
- Collected and analyzed user responses to guide feature prioritization and platform improvements.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, C, C++, SQL, Ruby
Frameworks & Libraries: React, Next, Node, Express, FastAPI, OpenCV, Tailwind CSS
Developer Tools: Git, GitHub, Docker, PostgreSQL, MongoDB, NPM, Bash, VS Code, BitBucket, Linux, Ubuntu
Concepts: CI/CD, RESTful APIs, Version Control, Backend Development, Databases, Object-Oriented Design